Reg. No. | | | | | | | | | | | [14MA11]

FIRST SEMESTER B.TECH EXAMINATION, DECEMBER – 2015

MATHEMATICS-I

Time: 3 Hours **Maximum Marks:70**

Answer any Five full questions. Select atleast TWO from each part.

PART –A

1. a. A body in air at 25°C cools from 100°C to 75°C in 1 minute. Find the temperature of the body at the end of 3 minutes? **04**

b. The number N of the bacteria in a culture grew at at rate proportional to N. The value of N was initially 100 and increased to 332 in 1 hour. What was the value of N after 1.5 hours? **05**

c. A series circuit with resistance R, inductance L and electromotive force E is governed by the differential equation $L\frac{di}{dt}+Ri=E$, where L and R are constants and initially the current i is zero. Find the current at any time t? **05**

2. a. Solve: $\frac{dy}{dx}=\frac{x(2\log x+1)}{\sin y+y\cos y}$. **04**

b. Solve: $\left(4xy+3y^2-x\right)dx+x(x+2y)dy=0$. **05**

c. Solve: $xy(1+xy^2)\frac{dy}{dx}=1$. **05**

3. a. Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2}+\frac{y^2}{b^2+\lambda}=1$, where λ is the parameter. **04**

b. Show that the family of parabolas $y^2=4a(x+a)$ is self-orthogonal. **05**

c. Find the orthogonal trajectories of the family of curves $r^n=a^n\cos n\theta$. **05**

4. a. Solve $\frac{d^2x}{dt^2}+4\frac{dx}{dt}+29x=0$. **04**

b. Solve $\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=2x+x^2$. **05**

c. Solve: $\left(D^2-2D+5\right)y=e^{2x}\sin x$. **05**

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PART-B

5. a. Solve $\frac{d^2y}{dx^2} + y = \tan x$ by the method of variation of parameters. **04**

b. Solve: $x^2y'' - xy' + 2y = x\sin(\log x)$. **05**

c. Solve: $(1+x)^2 y'' + (1+x)y' + y = \sin 2[\log(1+x)]$. **05**

6. a. Find the Laplace transform of Cos(at) by using the definition. **04**

b. Find the Laplace transform of the function t^2Sinat. **05**

c. Find the Laplace transform of the full wave rectifier $f(t) = E\sin wt,\ 0 < t < \pi/w$ having period π/w. **05**

7. a. Find the inverse Laplace transform of $\frac{s+2}{s^2(s+3)}$. **04**

b. Apply convolution theorem to evaluate $L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right]$. **05**

c. Use Laplace transform method to solve $y''(t) + 4y'(t) + 4y(t) = e^{-t}$ with y=0 y′=0 at t=0. **05**

8. a. Find all Eigen values Eigen vectors of the matrix, $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$. **04**

b. Solve the simultaneous equations $\frac{dx}{dt} + 2y + \sin t = 0$, $\frac{dy}{dt} - 2x - \cos t = 0$ given that x=0 and y=1 when t=0. **05**

c. Diagonalize the matrix $\begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$. **05**